

Optimizing Lean Transformers for Low-Resource Korean Chatbots: An 8-Case Ablation and Attention Analysis

Seongyeon Hwang
AIFEL Course Researcher
hwang@aifell.edu

Abstract

Deploying deep Transformer models in low-resource environments (e.g., small datasets with $\approx 10k$ samples) poses challenges due to overfitting and computational overhead. In this paper, we propose a “Lean Transformer”—a lightweight, 1-layer encoder-decoder architecture optimized for low-resource Korean dialogue generation. To analyze the architectural trade-offs in this low-resource regime, we design and conduct a controlled 8-case ablation study controlling three independent variables: embedding initialization (Scratch vs. Pre-trained Word2Vec), parameter weight-tying (Shared vs. Individual), and lexical substitution-based data augmentation.

Our quantitative results demonstrate that while the pre-trained, individual configuration (Case 3) achieves a BLEU-4 score of 14.38% (Beam Search, $K = 3$) with a validation loss of 3.1776, the scratch, weight-tied configuration (Case 2) achieves a BLEU-4 score of 14.37% while reducing trainable parameters by approximately 68% (from 6.4 million to 2.0 million). Conversely, the fully integrated configuration (Case 8), which combines pre-trained embeddings, parameter sharing, and data augmentation, shows a lower BLEU-4 score of 11.46%, exhibiting over-regularization constraints.

Furthermore, qualitative dialogue evaluation shows that Case 2 produces contextually aligned responses compared to Case 3, despite the marginal difference in BLEU-4. Explainable AI (XAI) analysis via cross-attention maps corroborates these findings, revealing that the shallow 1-layer structure is prone to “Attention Sinks” and “Vertical Banding” patterns when over-regularized, which degrades dynamic sequence alignment. Our findings provide a practical blueprint for developing parameter-efficient, low-resource dialogue agents.

1. Introduction

While Large Language Models (LLMs) excel at natural language generation, their computational footprints render deployment on edge devices or in resource-constrained environments challenging. Furthermore, specialized domains often suffer from data scarcity, typically providing datasets of around 10,000 samples. Under such low-resource constraints, standard deep Transformers are prone to overfitting and numerical instability due to over-parameterization.

To address these limitations, we evaluate the structural limits of model miniaturization by proposing a 1-layer encoder-decoder “Lean Transformer.” We investigate whether a compact architecture can achieve effective dialogue generation through an 8-case ablation study. We control three key independent variables: embedding initialization (Scratch vs. Pre-trained `ko.bin`), weight-tying (Individual vs. 3-Way Shared), and data augmentation (OFF vs. ON via lexical synonym substitution).

Our study makes three primary contributions:

1. We demonstrate that a compact, parameter-tied scratch configuration (Case 2) approaches the performance (14.37%) of a larger pretrained individual model (Case 3) while reducing trainable parameters by 68% (from 6.4M to 2.0M).
2. We identify a quantitative limit in the fully integrated configuration (Case 8), demonstrating that combining pre-trained embeddings, parameter sharing, and synonym augmentation results in over-regularization.
3. We leverage cross-attention maps to visualize the structural limits of 1-layer models, characterizing the emergence of “Attention Sinks” and “Vertical Banding” patterns under parameter constraints.

2. Related Work

2.1. Low-Resource Neural Dialogue Generation

Standard deep networks are susceptible to overfitting when the training corpus is small (e.g., fewer than 10,000 conversational pairs), often memorizing the limited sequences

instead of learning the underlying semantic mapping. Prior literature has attempted to mitigate this by transferring representations from pre-trained language models. However, direct fine-tuning of large models in resource-constrained environments (such as mobile processors or edge devices) remains constrained due to computational overhead. This highlights the importance of studying compact, single-layer networks that minimize parameter redundancy while maintaining generalization capabilities.

2.2. Weight-Tying in Transformer Architectures

Weight-tying (or parameter sharing) is a technique in neural language modeling and machine translation to reduce the model’s footprint and act as a regularizer [1, 2]. This sharing forces the encoder embedding matrix, the decoder embedding matrix, and the final output linear projection layer to share the exact same weight parameters. In large-scale systems, this sharing reduces memory footprint and regularizes the shared vector space. However, the majority of existing research evaluates weight-tying in deep configurations (e.g., 6 or more layers). Its impact on shallow, 1-layer models remains to be fully analyzed, raising the question of whether the reduction in parameter capacity (from 6.4 million to 2.0 million in our setup) acts as a regularizer or a bottleneck that limits the model’s expressive capacity.

2.3. Lexical Substitution Data Augmentation in Morphological Languages

To combat data scarcity, data augmentation has become a standard approach to artificial corpus expansion. A popular technique is Lexical Substitution (Synonym Replacement), where words in the training sequence are randomly replaced by synonyms retrieved from thesauri or computed via cosine similarity in static word embedding spaces. While this approach expands vocabulary coverage, its application to morphologically complex languages like Korean presents structural challenges. Korean is an agglutinative language where content words (nouns, verb stems) are bound to various particles, postpositions, and suffixes (known as *josa* and *eomi*). Synonym substitution based on vector space distance can replace words with contextually inconsistent synonyms or antonyms. This introduces syntactic and semantic noise into the parallel corpus, which can disrupt the alignment mechanisms of single-layer attention heads.

3. Proposed Methodology

3.1. Lean Transformer Architecture Specifications

We implement a compact, 1-layer encoder-decoder Transformer architecture (the “Lean Transformer”). Rather than relying on standard deep structures, we restrict the layers to $N = 1$ for both the encoder and decoder. The model hyperparameters are controlled and fixed across all experimental

configurations to ensure a controlled ablation study:

- **Embedding and Hidden Dimension (d_{model}):** Fixed at 200 to maintain structural compatibility with the pre-trained static word embeddings.
- **Feed-Forward Network Hidden Dimension (d_{ff}):** Fixed at 1024, which provides a wide bottleneck for information routing.
- **Attention Heads (h):** Fixed at 4. This divides the hidden state into head dimensions of $d_k = d_{model}/h = 50$.
- **Dropout and Regularization:** A dropout rate of 0.2 is applied to all sub-layers (attention outputs, feed-forward outputs, and positional encodings) to provide a regularization layer in the shallow network.

3.2. Embedding Initialization and Variance Calibration

By default, the `nn.Embedding` module in PyTorch initializes weights with a standard deviation of $\sigma \approx 1.0$. However, when parameter weight-tying is enabled, the embedding weight matrix is shared with the final fully connected (FC) output projection layer. Under this configuration, if the projection weights have a standard deviation of $\sigma \approx 1.0$, their matrix multiplication with the decoder hidden states yields disproportionately large logits. This causes logit explosion, leading to Softmax saturation, gradient instability, and cross-entropy loss values exceeding 800 in early training epochs.

To prevent logit explosion under weight-tying, we apply a variance calibration strategy at the initialization stage for both Scratch and Pre-trained configurations:

1. **Scratch Embeddings:** The weights are initialized using a normal distribution with a mean of 0.0 and a standard deviation scaled to the inverse square root of the model dimension:

$$\mathbf{W}_{scratch} \sim \mathcal{N}\left(0, \frac{1}{d_{model}}\right) \quad (1)$$

For $d_{model} = 200$, this constrains the initial weight variance to $\sigma = 1/\sqrt{200} \approx 0.07$.

2. **Pre-trained Embeddings:** The static Korean Word2Vec embeddings (`ko.bin`), which initially possess a standard deviation of $\sigma \approx 1.0$, are scaled down by dividing them by $\sqrt{d_{model}}$ before being copied into the embedding layer:

$$\mathbf{W}_{pretrained} = \frac{\mathbf{W}_{ko.bin}}{\sqrt{d_{model}}} \quad (2)$$

This division scales the pre-trained weights to match the same standard deviation ($\sigma \approx 0.07$) as the scratch setup.

During the forward pass, both configurations utilize the exact same scaling lookup and sequence mapping logic. The looked-up vector is multiplied by $\sqrt{d_{model}}$ to scale its variance back to ≈ 1.0 , ensuring it mixes proportionally with

Table 1. 8-Case Ablation Study Matrix and Experimental Objectives.

Experimental ID	Embedding (E)	Weight-Tying (W)	Augmentation (A)	Core Objective and Proof Target
Case 1	Scratch	Individual	OFF	Absolute Baseline: Control group with zero structural optimization.
Case 2	Scratch	Shared (Tied)	OFF	Parameter Sharing: Evaluates 3-way weight tying in a scratch setup.
Case 3	Pre-trained	Individual	OFF	Prior Knowledge: Evaluates pre-trained embedding initialization.
Case 4	Pre-trained	Shared (Tied)	OFF	Initial Synergy: Combines weight-tying and pre-trained embeddings.
Case 5	Scratch	Individual	ON	Data Scaling: Evaluates lexical data augmentation.
Case 6	Scratch	Shared (Tied)	ON	Augmented Sharing: Evaluates parameter sharing in an augmented data setup.
Case 7	Pre-trained	Individual	ON	Knowledge and Data: Combines pre-trained embeddings and data augmentation.
Case 8	Pre-trained	Shared (Tied)	ON	Fully Integrated: Combines all three techniques to check synergies.

ited variance is attributed to the regularizing effect of the Label Smoothing = 0.1 constraint applied to the cross-entropy target distributions. By preventing the network from assigning full 1.0 confidence to target tokens, label smoothing establishes a loss floor, stabilizing training while narrowing the variance between configurations.

2. **Augmentation-Driven Convergence Acceleration:** Models trained on augmented data (Cases 5–8) undergo early stopping at lower epoch counts (e.g., converging at epoch 4 for Cases 5 and 7) compared to non-augmented baseline models (e.g., Case 2 converged at epoch 15). This is explained by two factors. First, the 2.13x expansion of training sequences increases the number of parameter updates (steps) per epoch from 164 to 349, accelerating convergence in terms of epoch metrics. Second, the lexical redundancy introduced by synonym substitution leads to early saturation of semantic generalization, prompting the early stopping criteria to trigger early.

4.3. BLEU-4 Performance and Ablation Analysis

To assess the generative quality of the chatbot, we measure the BLEU-4 score (4-gram exact text overlap) against the validation target set. Table 2 summarizes the quantitative results under both Greedy decoding and Beam Search ($K = 3$) configurations.

Analyzing the results reveals crucial structural interactions:

- **The Efficacy of Parameter Sharing (Case 2):** In a scratch-initialized setup, applying 3-way parameter weight-tying (Case 2) increases BLEU-4 from 12.89% (Case 1) to 14.37%. This shows that in small-data environments, parameter reduction acts as a regularizer, con-

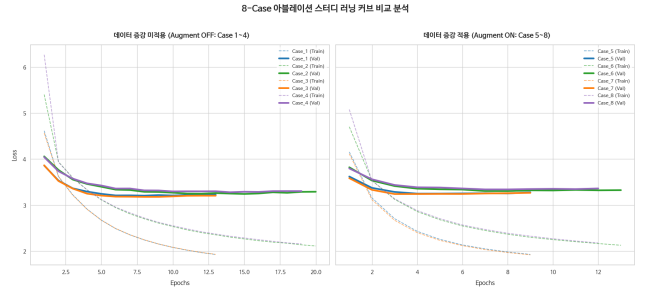


Figure 1. Training loss convergence curves across the 8 ablation cases. All configurations converge to a narrow validation loss band (3.18–3.34) due to label smoothing. Augmented models (Cases 5–8) reach early stopping at lower epoch counts than non-augmented baselines.

straining the 1-layer network’s hypothesis space and preventing overfitting.

- **The Synergy and Limit of Pre-training (Case 3 vs. Case 4):** Case 3 (Pre-trained + Individual) achieves the highest BLEU-4 scores (Val Loss of 3.1776, Beam BLEU of 14.38%), illustrating the value of external semantic knowledge. However, when parameter sharing is introduced alongside pre-trained embeddings (Case 4), BLEU-4 drops to 13.70%, indicating a capacity constraint. Forcing a pre-trained embedding space to be shared across both encoding (input comprehension) and decoding (output projection) in a shallow 1-layer model restricts its representational freedom, compressing the shared vector space and degrading generation.
- **The Impact of Fully Integrated Regularization (Case 8):** Applying all three methods simultaneously (Case 8) leads to the lowest score in the entire ablation matrix (Beam BLEU of 11.46%, Val Loss of 3.3394), indicating

Table 2. Quantitative results across the 8-case ablation study matrix. Convergence Epoch is reported as *best_epoch* / *stop_epoch*, where *best_epoch* denotes the epoch with the lowest validation loss and *stop_epoch* denotes the epoch at which early stopping was triggered (*patience* = 5). Case 2 and Case 3 achieve comparable Beam BLEU-4 scores (14.37% vs. 14.38%), while Case 8 records the lowest score (11.46%).

ID	Embedding (E)	Weight Sharing (W)	Augmentation (A)	Best Val Loss	Convergence Epoch	Greedy BLEU-4 (%)	Beam BLEU-4 ($K = 3$) (%)
Case 1	Scratch	Individual	OFF	3.2044	8 / 13	11.60	12.89
Case 2	Scratch	Shared (Tied)	OFF	3.2411	15 / 20	12.64	14.37
Case 3	Pre-trained	Individual	OFF	3.1776	8 / 13	12.73	14.38
Case 4	Pre-trained	Shared (Tied)	OFF	3.2777	14 / 19	11.91	13.70
Case 5	Scratch	Individual	ON	3.2481	4 / 9	10.68	11.92
Case 6	Scratch	Shared (Tied)	ON	3.3026	8 / 13	11.46	12.10
Case 7	Pre-trained	Individual	ON	3.2395	4 / 9	11.28	12.32
Case 8	Pre-trained	Shared (Tied)	ON	3.3394	7 / 12	10.29	11.46

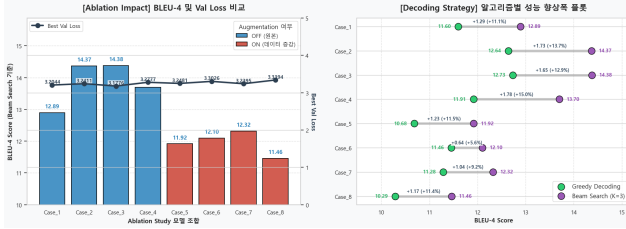


Figure 2. Validation BLEU-4 scores across the 8 ablation configurations. (Left) Absolute scores for Greedy and Beam Search ($K = 3$). (Right) Dumbbell plot showing per-case Beam Search gains, which range from +0.64%p (Case 6) to +1.78%p (Case 4), indicating that multi-path decoding consistently compensates for the shallow architecture’s flat output distributions.

a capacity limitation. When the shallow 1-layer model is constrained by pre-trained constraints, parameter sharing, and synonym data noise, it shows limited capacity to map dialogue contexts.

4.4. Influence of Decoding Strategy

As illustrated by the dumbbell plots in Figure 2, utilizing Beam Search ($K = 3$) yields an increase over Greedy decoding across all eight configurations. The BLEU-4 gains range from +0.64% (Case 6) to +1.78% (Case 4), representing relative improvements of up to 15.0%. These results show that multiple path search extracts correct tokens from flat vocabulary probability distributions, compensating for the representational capacity of shallow 1-layer networks during inference.

5. Qualitative Dialogue and Attention Analysis

While automatic metrics such as BLEU-4 and cross-entropy validation loss provide a convenient quantitative overview of model performance, they fail to capture the dialogue generation quality. A chatbot’s quality is defined by its ability to engage in coherent, grammatically stable, and contextually aligned conversations. In this section, we analyze conversational quality alongside statistical benchmarks and

link these generated outputs to the inner mathematical states of the models via cross-attention map analysis (Explainable AI).

5.1. Qualitative Evaluation Dimensions

To assess the generated responses, we establish three qualitative evaluation dimensions:

- Contextual Alignment:** The model’s ability to maintain topic consistency and correctly interpret the user’s underlying intent (e.g., distinguishing between separation grief, romantic excitement, and routine daily descriptions).
- Grammatical Integrity:** The structural completeness and morpho-syntactic correctness of the generated Korean sentence, ensuring no missing particles or unnatural morphemic splices.
- Empathetic Resonance:** The model’s ability to formulate emotionally supportive and deep responses that convey empathy, validation, and comfort, rather than relying on generic, repetitive answers.

5.2. Comparative Analysis of Dialogue Generations

To highlight the impact of the proposed architectural variants, we extract representative dialogue responses generated by the quantitative top-performing configurations—Case 2 (Scratch + Shared, BLEU: 14.37

Analyzing Table 3 reveals structural differences between the models:

- Case 2 (Scratch + Shared):** Despite having a minor 0.01
- Case 3 (Pre-trained + Individual):** While grammatically stable for standard prompts (*“Have confidence”*), Case 3 generates repetitive outputs (*“Your mind must be complicated”* generated across multiple separate queries) and context mismatches. When presented with the emotional query (*“It feels strange walking the path alone...”*), Case 3 outputs a literal, shallow response: *“I should take a walk.”* This represents a shallow keyword-association pattern.
- Case 8 (Pre-trained + Shared + Augmented):** Case 8 exhibits semantic inversion and mechanical tautology. For

Table 3. Qualitative comparison of dialogue generation across daily life, separation, and romance domains. Case 2 (Scratch + Shared) demonstrates stronger contextual alignment and empathetic resonance, whereas Case 3 (Pre-trained + Individual) exhibits keyword-level shallow matching and Case 8 (Fully Integrated) shows semantic polarity inversion and tautological repetition. Annotation tags (bold) summarize the observed linguistic phenomena.

Domain	User Input (Query)	Case 2 (Scratch + Shared)	Case 3 (Pre-trained + Individual)	Case 8 (Pre-trained + Shared + Aug)
Daily Life	여름에는 냉면만한게 없는 듯 *(Nothing beats naengmyeon)*	생각만 하어도 군침이 돌네요. *(My mouth waters just thinking...)*	만 하어도 군침이 돌네요. *(...waters just thinking...) (Morpheme Omission)	생각만 하어도 괜찮다고 생각해보세요. *(Try thinking it's okay just...) (Tautological Redundancy)
Separation	이제 다 잊었다고 생각했는데 다시 생각나니까 힘들어. *(I thought I forgot, but...)	쉽지 않은 결정이었을 터인데 안타깝네요. *(It must have been a hard decision...) (Deep Empathetic Validation)	마음이 복잡하겠어요. *(Your mind must be complicated.) (Repetitive / Shallow Response)	언젠간 잊히어 지르게 이예요. *(It will be forgotten someday.) (Detached / Standard Advice)
Separation	항상 같이 걷던 길을 혼자 걸으니깐 기분이 이상해. *(It feels strange walking...)	혼자서도 잘하는 연습이 필요하어요. *(You need practice to do well...) (Constructive Encouragement)	산책 좀 해야겠네요. *(I should take a walk.) (Context Mismatch)	자신에게 돌아가는 것이 좋겠네요. *(It would be good to return...) (Semantic Coherence Breakdown)
Romance	마음에 드는 사람이 있는데 *(There is someone I like)*	저어도 간절히 기도하세요. *(I will pray earnestly for you...) (Warm Emotional Support)	자신감을 가지세요. *(Have confidence.) (Generic Solitary Advice)	마음이 아프겠어요. *(Your heart must be hurting.) (Semantic Polarity Inversion)

the romance prompt (*“There is someone I like”*), it outputs *‘‘Your heart must be hurting,’’* demonstrating an inversion of emotional polarity. It also generates repetitive structures (*‘‘Try thinking it’s okay just thinking about it’’*), indicating over-regularization.

5.3. Qualitative Cross-Attention Map Observations

To examine these responses, we analyze the inner states of Case 2, Case 3, and Case 8 by extracting their Max-Pooled cross-attention maps for the emotional query: “항상 같이 걷던 길을 혼자 걸으니깐 기분이 이상해.” *(It feels strange walking the path alone we always walked together.)*. Figure 3 illustrates these attention mapping patterns.

A visual inspection of the heatmaps reveals three distinct attention routing behaviors:

1. **Case 2 (Sparse Local Alignment):** In Case 2, the attention map shows distinct, sparse attention weights. The model maps output tokens to crucial semantic pivots in the input, routing attention weights to the emotional stems “혼자” *(alone)* and “기분이 이상해” *(feels

strange)*. This localized attention corresponds to the contextually rich advice: *‘‘You need practice to do well alone.’’*

2. **Case 3 (Shallow Association / Attention Sink):** In Case 3, the attention weight is concentrated on the early tokens—“항상 같이 걷던” *(always walked together)*—showing lower weights at the end of the query (“기분이 이상해”). This concentration represents a shallow keyword-association pattern: the model relies on static surface-level keywords instead of dynamic sequence mapping, resulting in the output: *‘‘I should take a walk.’’*
3. **Case 8 (Vertical Banding Anomaly):** In Case 8, the cross-attention map exhibits vertical alignment (vertical banding), indicating that attention is distributed uniformly across specific indexes across almost all decoding steps rather than dynamically across the sequence. This represents an “Attention Sink” pattern: under multiple regularizing constraints and lexical noise, the model fails to perform dynamic input-output alignment, instead distributing weights uniformly across specific index po-

Target Query: "내일 좋아하는 사람이라 처음으로 단들이 밥 먹기로 했어." (All Heads & Max Pooled)

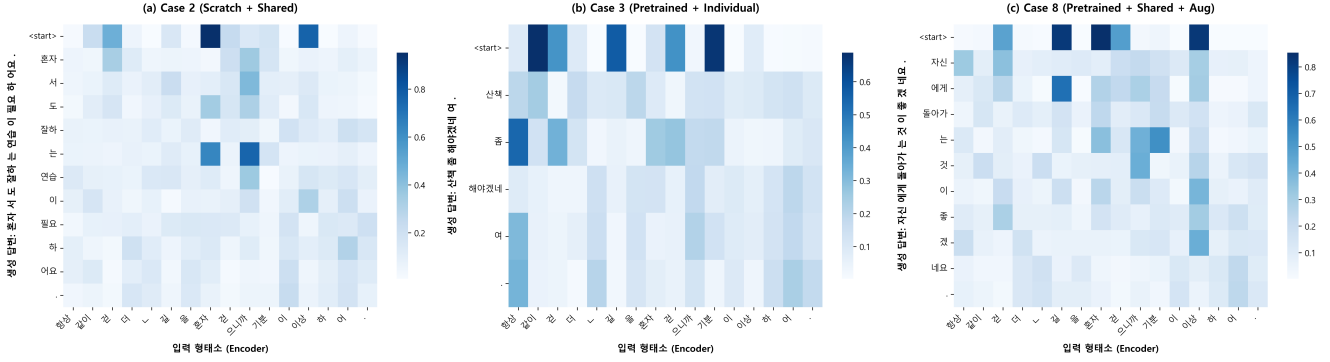


Figure 3. Max-pooled cross-attention heatmaps for the query “항상 같이 걸던 길을 혼자 걸으니까 기분이 이상해” (*It feels strange walking the path alone we always walked together*). (Left) Case 2 exhibits sparse, localized attention on the emotional pivots “혼자” and “기분이 이상해.” (Middle) Case 3 concentrates attention on early tokens (“항상 같이 걸던”), neglecting the emotional suffix. (Right) Case 8 shows vertical banding, where attention is uniformly locked onto fixed index positions across decoding steps.

sitions.

6. Conclusion and Future Work

6.1. Study Summary and Key Contributions

In this study, we investigated the structural limits of model miniaturization by evaluating a 1-layer encoder-decoder “Lean Transformer” on a low-resource Korean dialogue corpus. Through a controlled 8-case ablation study, we isolated the individual and combined effects of embedding initialization (E), parameter weight-tying (W), and lexical data augmentation (A).

Our primary empirical finding is that **Case 2 (Scratch + Shared)** represents a parameter-efficient setup. Case 2 reduces trainable parameters by 68% (from 6.4 million to 2.0 million weights) while achieving comparable quantitative performance to the larger Case 3 model (Case 2’s 14.37% vs. Case 3’s 14.38% BLEU-4). Qualitative and cross-attention (XAI) diagnostic evaluations indicate that Case 2 maintains contextual consistency and dynamic attention mapping more effectively than the individual pre-trained configuration (Case 3). Conversely, combining all three techniques (Case 8) leads to over-regularization constraints, manifesting as vertical banding anomalies in cross-attention heatmaps due to capacity bottlenecks.

6.2. Discussion on Automated Overlap Metrics under One-to-Many Distributions

The quantitative results, particularly the relatively low BLEU-4 scores (ranging from 10.29% to 14.38%) across all configurations, highlight the mathematical behavior of exact-match overlap metrics in dialogue systems rather than a lack of conversational capability.

Unlike machine translation or summarization where the

target semantic space is relatively constrained, dialogue generation inherently exhibits a “One-to-Many” mapping property, where a single query can be answered by multiple semantically valid but lexically diverse responses. Under this setting, even if a model generates a response that is grammatically correct and contextually appropriate, it receives a lower score under N-gram matching metrics like BLEU-4 if the tokens do not match the single ground-truth reference sequence.

This mathematical behavior explains the performance trends observed in our ablation study. For instance, although Case 2 generates highly coherent and contextually aligned dialogues, its quantitative score remains statistically comparable to or slightly below Case 3. Furthermore, this behavior clarifies why the data-augmented models (Cases 5–8) exhibit lower BLEU-4 scores. While lexical synonym substitution during training expands the model’s vocabulary and generalization capability under morphological variations, it simultaneously increases the probability that the generated response will deviate from the specific literal tokens of the single validation reference. As a result, the augmented models suffer a statistical penalty in string-matching benchmarks, despite generating semantically viable dialogue.

These findings suggest that automated overlap metrics have inherent limitations as primary optimization targets in low-resource dialogue research. Consequently, they should be balanced with qualitative evaluations and cross-attention diagnostics to ensure accurate architectural assessment.

6.3. Research Limitations and Future Directions

1. **Linguistic Noise in Static Synonym Substitution:** Lexical substitution via static Word2Vec similarity replaced tokens with contextually inconsistent synonyms or antonyms due to the morphologically complex nature

of Korean. Future work will replace Word2Vec synonym replacement with Masked Language Modeling (MLM) approaches leveraging context-aware bidirectional models (e.g., KoBERT or RoBERTa) to generate semantically preserved augmented samples.

2. **Structural Sinks in Shallow Attention:** The single-layer attention mechanism is vulnerable to vertical banding noise when constrained. To stack deeper layers (e.g., 2 or 3 layers) on small datasets without triggering overfitting, future experiments will evaluate advanced regularization techniques such as DropConnect, Layer-Dropout, and stochastic depth training.

References

- [1] H. Inan, K. Khosravi, and R. Socher. 2017. Tying word embeddings and language model weights: A loss framework. In *Proceedings of the 5th International Conference on Learning Representations (ICLR 2017)*. [2](#)
- [2] O. Press and L. Wolf. 2017. Using the output embedding to improve language models. In *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2017)*, Volume 2: Short Papers, pages 157–163. [2](#)
- [3] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin. 2017. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS 2017)*, 30, pages 5998–6008.